

ELEVATE LABS TASK

9

Network Vulnerability Scanning

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Domain: Cybersecurity

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1. Scan local network.

Local network scanning means discovering devices (hosts) connected to the same network (LAN) as your system.

Why we scan a local network

- Identify **connected devices** (PCs, mobiles, routers, servers)
- Find **live IP addresses** • Detect **open ports & services**
- Used in:
 - Network administration
 - Ethical hacking & penetration testing
 - Troubleshooting network issues

Step 1: Install Nmap sudo

apt install nmap

```
(kali@kali)-[/]
└─$ sudo apt install nmap
nmap is already the newest version (7.98+dfsg-1kali1).
nmap set to manually installed.
The following packages were automatically installed and are no longer required:
  anass-common libavformat61 libgdal37 libjs-underscore libobjc-14-dev libsqlcipher1 mesa-vaapi-drivers python3-kismetcapture
  bloodhound.py libbluray2 libgeos3.14.0 libmjpegutils-2.1-0t64 libplacebo349 libswscale8 pocketsphinx-en-us python3-kismetcapture
  curlftpfs libbson-1.0-0t64 libgirepository-1.0-1 libmongoc-1.0-0t64 libpocketsphinx3 libudfread0 python3-bluepy python3-kismetcapture
  gir1.2-girepository-2.0 libconfig-inifiles-perl libpgmell1t64 libmpeg2encpp-2.1-0t64 libportmidi0 libvdpau-va-gl1 python3-click-plugins python3-kismetcapture
  libarmadillo14 libdisplay-info2 libpgmep6t64 libplex2-2.1-0t64 libpostproc58 libwirespark18 python3-fs python3-kismetcapture
  libaudio2 libfuse2t64 libinstpatch-1.0-2 libmupdf25.1 libradare2-5.0-0t64 libwiretap15 python3-gpg python3-kismetcapture
  libavfilter10 libgavl-1 libjs-jquery-ui libnet1 libsphinxbase3t64 libwsutil6 python3-kismetcapturebtgeiger python3-xmlutils

Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 15
```

Step 2: Scan for Live Hosts (Ping Scan) nmap

-sn ip address

```
(kali㉿kali)-[/]
$ nmap -sn 192.168.10.3
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-31 10:01 -0500
Nmap scan report for 192.168.10.3
Host is up.
Nmap done: 1 IP address (1 host up) scanned in 0.59 seconds
```

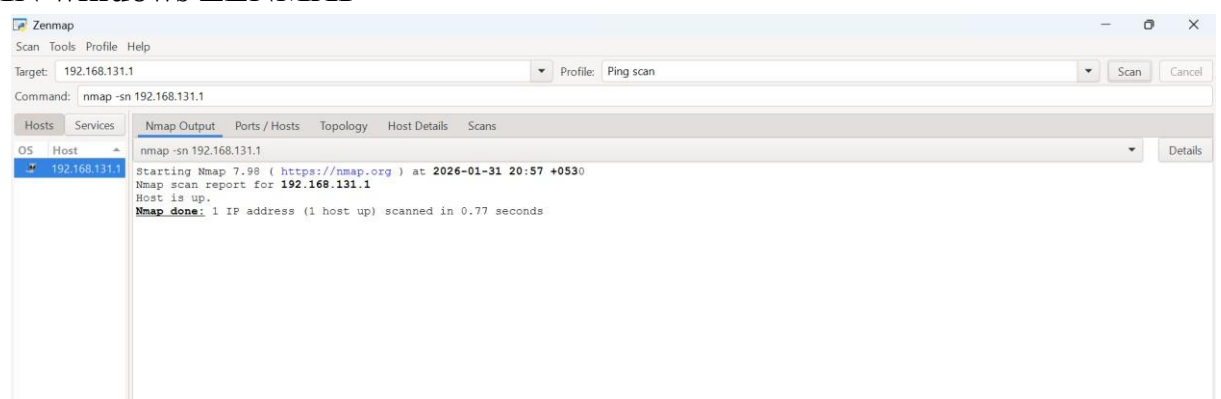
Practical 2: Scan Open Ports on a Specific Host

```
(kali㉿kali)-[/]
$ nmap 192.168.10.3
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-31 10:04 -0500
Nmap scan report for 192.168.10.3
Host is up (0.00015s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
```

Practical 3: Scan All Ports (Advanced

```
(kali㉿kali)-[/]
$ nmap -p- 192.168.10.3
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-31 10:05 -0500
Stats: 0:01:51 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 77.15% done; ETC: 10:08 (0:00:33 remaining)
Nmap scan report for 192.168.10.3
Host is up (0.00015s latency).
Not shown: 65534 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
```

IN windows ZENMAP



2. Identify open ports

A port is a logical communication endpoint.

Each service listens on a port number.

Why Identify Open Ports?

- Discover running services
- Find attack surface
- First step in penetration testing
- Used in network administration

Step 1: Basic Open Port Scan

```
(kali㉿kali)-[~]  
$ nmap 192.168.10.3  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-31 10:35 -0500  
Nmap scan report for 192.168.10.3  
Host is up (0.000039s latency).  
Not shown: 999 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
80/tcp    open  http  
  
Nmap done: 1 IP address (1 host up) scanned in 7.16 seconds
```

Step 2: Scan Specific Ports nmap

-p 22,80,443 ip address

```
(kali㉿kali)-[~]  
$ nmap -p 22,80,443 192.168.10.3  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-31 10:37 -0500  
Nmap scan report for 192.168.10.3  
Host is up (0.00018s latency).  
  
PORT      STATE SERVICE  
22/tcp    filtered ssh  
80/tcp    open  http  
443/tcp   filtered https
```

Step 3: Scan All Ports

```
(kali㉿kali)-[~]  
$ nmap -p- 192.168.10.3  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-31 10:38 -0500  
Stats: 0:01:08 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 56.47% done; ETC: 10:40 (0:00:52 remaining)  
Stats: 0:01:33 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 86.33% done; ETC: 10:40 (0:00:15 remaining)  
Nmap scan report for 192.168.10.3  
Host is up (0.000061s latency).  
Not shown: 65534 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
80/tcp    open  http
```

Step 4: Service Version Detection nmap

-Sv

```
(kali@kali)-[~]
$ nmap -sV 192.168.10.3
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-31 10:41 -0500
Nmap scan report for 192.168.10.3
Host is up (0.000029s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
80/tcp    open  http      Apache httpd 2.4.66 ((Debian))

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.16 seconds
```

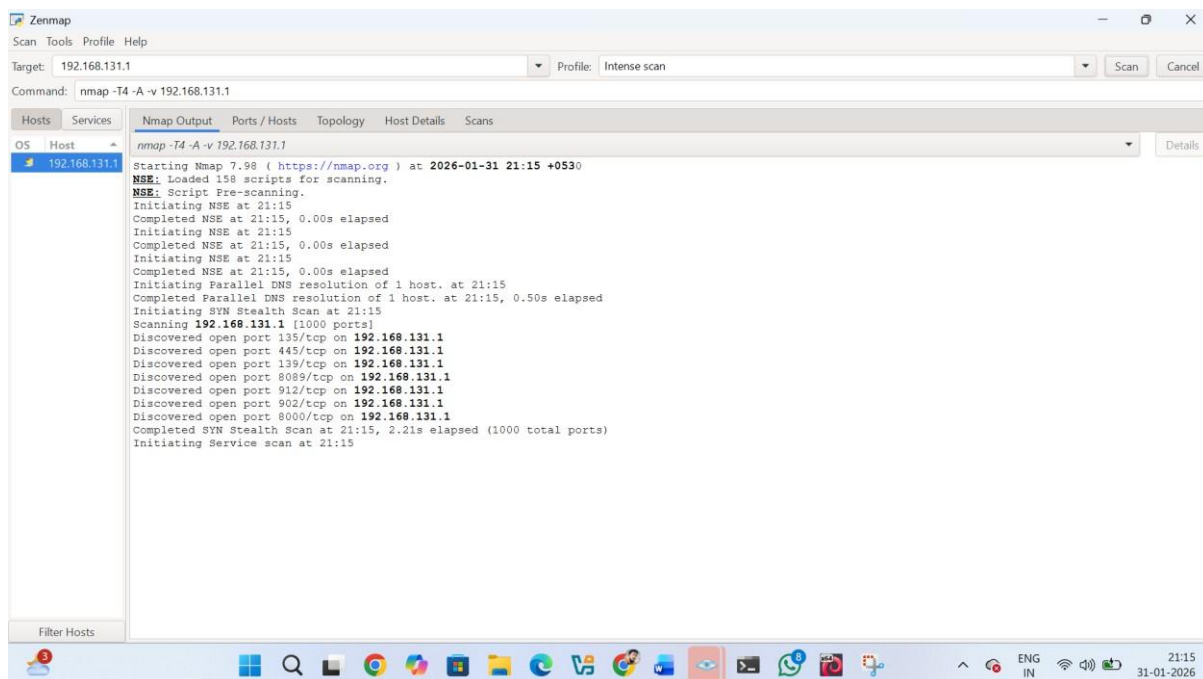
Step 5: Aggressive Scan nmap

-A

```
(kali@kali)-[~]
$ nmap -A 192.168.10.3
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-31 10:43 -0500
Nmap scan report for 192.168.10.3
Host is up (0.000083s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
80/tcp    open  http      Apache httpd 2.4.66 ((Debian))
|_http-server-header: Apache/2.4.66 (Debian)
|_http-title: Apache2 Debian Default Page: It works
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose
Running: Linux 5.X|6.X
OS CPE: cpe:/o:linux:linux_kernel:5 cpe:/o:linux:linux_kernel:6
OS details: Linux 5.0 - 6.2
Network Distance: 0 hops

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 16.67 seconds
```

Windows



3. Detect services

Service detection is the process of identifying the exact services and their versions running on open ports of a system. When a port is found open, it only indicates that some application is listening; service detection goes a step further by determining *what* application it is, such as Apache web server, OpenSSH, FTP, or MySQL, and often its version number. Tools like Nmap perform service detection by sending specially crafted probes to open ports and analyzing the responses or banners returned by the target system. This information is very important in network security and ethical hacking because outdated or misconfigured services may contain known vulnerabilities. Service detection helps security professionals and system administrators understand the services exposed on a network, assess potential risks, and plan appropriate security measures or further testing.

Step 1: Identify Target IP

192.168.10.3

Step 2: Basic Service Detection

```
(kali@kali)-[~]
$ nmap -sV 192.168.10.3
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-31 10:51 -0500
Nmap scan report for 192.168.10.3
Host is up (0.00012s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
80/tcp    open  http    Apache httpd 2.4.66 ((Debian))

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 14.84 seconds
```

Step 3: Service Detection on Specific Ports nmap

-sV -p 22,80,3306

```
(kali@kali)-[~]
$ nmap -sV -p 22,80,3306 192.168.10.3
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-31 10:52 -0500
Nmap scan report for 192.168.10.3
Host is up (0.00019s latency).

PORT      STATE SERVICE VERSION
22/tcp    filtered ssh
80/tcp    open  http    Apache httpd 2.4.66 ((Debian))
3306/tcp  filtered mysql

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.91 seconds
```

Step 5: Increase Accuracy

nmap -sV --version-intensity 9

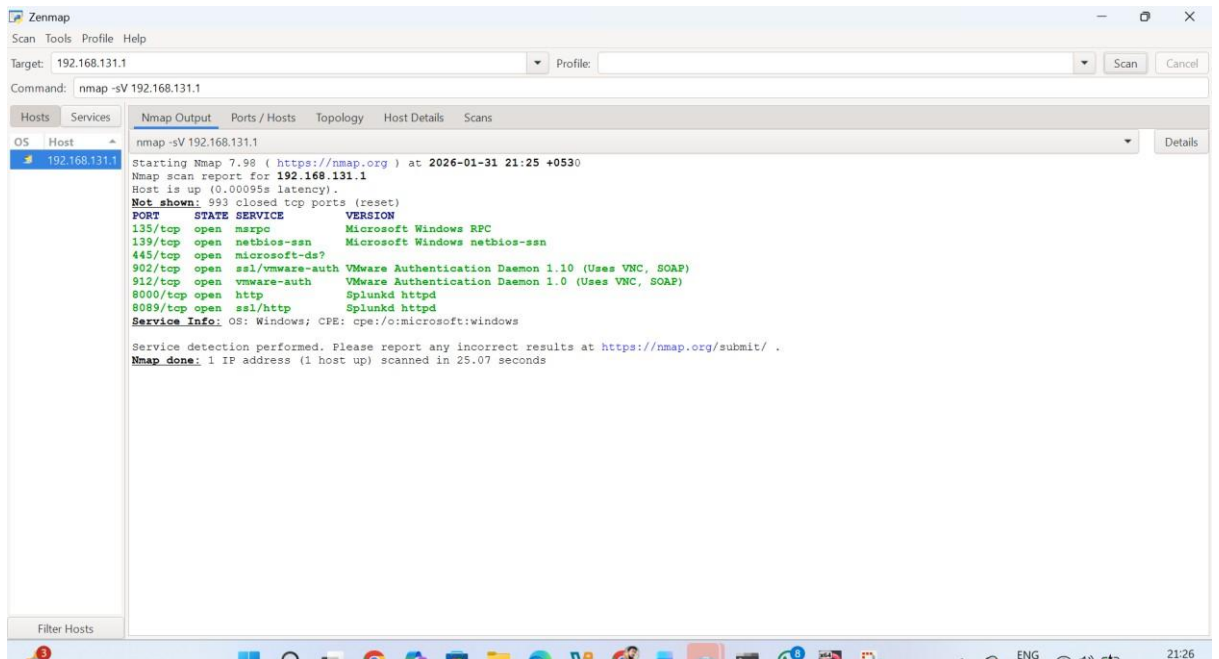

```

(kali@kali)-[~]
$ nmap -sV --version-intensity 9 192.168.10.3
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-31 10:54 -0500
Nmap scan report for 192.168.10.3
Host is up (0.000042s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
80/tcp    open  http      Apache httpd 2.4.66 ((Debian))

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 10.93 seconds

```

Windows:-



4. Identify OS :-

1. OS identification is also known as OS fingerprinting
2. It detects the type of operating system (Windows, Linux, Unix, etc.)
3. Uses TCP/IP stack behavior analysis
4. Helps in vulnerability assessment
5. Performed using tools like Nmap and ZenmaRequires at least one open and one closed port for accuracy

Step 1: Basic OS Detection sudo

nmap -O

```
(kali㉿kali)-[~]
$ sudo nmap -O 192.168.10.3
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-31 11:02 -0500
Nmap scan report for 192.168.10.3
Host is up (0.000092s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose
Running: Linux 5.X|6.X
OS CPE: cpe:/o:linux:linux_kernel:5 cpe:/o:linux:linux_kernel:6
OS details: Linux 5.0 - 6.2
Network Distance: 0 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.26 seconds
```

Step 2: OS Detection with Service Scan (Recommended) sudo

nmap -A

```
(kali㉿kali)-[~]
$ sudo nmap -A 192.168.10.3
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-31 11:03 -0500
Nmap scan report for 192.168.10.3
Host is up (0.00011s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
80/tcp    open  http      Apache httpd 2.4.66 ((Debian))
|_http-server-header: Apache/2.4.66 (Debian)
|_http-title: Apache2 Debian Default Page: It works
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose
Running: Linux 5.X|6.X
OS CPE: cpe:/o:linux:linux_kernel:5 cpe:/o:linux:linux_kernel:6
OS details: Linux 5.0 - 6.2
Network Distance: 0 hops

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 14.92 seconds
```

Step 3: Guess OS if Detection is Incomplete

sudo nmap -O --osscan-guess

```
(kali㉿kali)-[~]
$ sudo nmap -O --osscan-guess 192.168.10.3
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-31 11:05 -0500
Nmap scan report for 192.168.10.3
Host is up (0.000080s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose
Running: Linux 5.X|6.X
OS CPE: cpe:/o:linux:linux_kernel:5 cpe:/o:linux:linux_kernel:6
OS details: Linux 5.0 - 6.2
Network Distance: 0 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.26 seconds
```

5. Analyze vulnerability

Vulnerability analysis is the process of identifying security weaknesses in a system that could be exploited by attackers. These weaknesses may exist due to outdated software, misconfigured services, weak authentication, or unpatched operating systems. After identifying open ports, services, and the operating system, vulnerability analysis correlates this information with known vulnerabilities from databases such as CVE. Tools like Nmap use built-in scripts to automatically check for common vulnerabilities in running services. Vulnerability analysis is a critical phase in ethical hacking and network security, as it helps organizations understand their security risks and apply appropriate countermeasures before attackers exploit them.

Step 1: Basic Vulnerability Scan

`sudo nmap --script vuln`

```
(kali㉿kali)-[~]
└─$ sudo nmap --script vuln 192.168.10.3
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-31 11:09 -0500
Nmap scan report for 192.168.10.3
Host is up (0.000053s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
|_ http-stored-xss: Couldn't find any stored XSS vulnerabilities.
|_ http-csrf: Couldn't find any CSRF vulnerabilities.
|_ http-dombased-xss: Couldn't find any DOM based XSS.
|_ http-enum:
|   /test.php: Test page
|   /info.php: Possible information file
|_  /server-status/: Potentially interesting folder

Nmap done: 1 IP address (1 host up) scanned in 37.84 seconds
```

Step 2: Vulnerability Scan on Specific Ports

`sudo nmap --script vuln -p 80,443`

```
(kali㉿kali)-[~]
└─$ sudo nmap --script vuln -p 80,443 192.168.10.3
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-31 11:11 -0500
Nmap scan report for 192.168.10.3
Host is up (0.000070s latency).

PORT      STATE SERVICE
80/tcp    open  http
|_ http-dombased-xss: Couldn't find any DOM based XSS.
|_ http-stored-xss: Couldn't find any stored XSS vulnerabilities.
|_ http-csrf: Couldn't find any CSRF vulnerabilities.
|_ http-enum:
|   /test.php: Test page
|   /info.php: Possible information file
|_  /server-status/: Potentially interesting folder
443/tcp    filtered https

Nmap done: 1 IP address (1 host up) scanned in 32.63 seconds
```


6. Save Scan Results

Saving scan results is the process of storing the output of network scans for future reference, analysis, and reporting. During penetration testing or network assessment, scans may generate large amounts of data such as open ports, detected services, operating systems, and vulnerabilities. Saving these results allows security professionals to review findings later, compare changes over time, and prepare security reports. Tools like Nmap support multiple output formats including normal text, XML, and HTML, making it easier to document results or integrate them with other security tools. Properly saved scan results are essential for evidence collection, auditing, and professional documentation.

7. Step 1: Save Scan Output in Normal Text Format nmap

ip address -oN scan.txt

```
(kali㉿kali)-[~]
└─$ nmap 192.168.10.3 -oN scan.txt

Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-31 11:18 -0500
Nmap scan report for 192.168.10.3
Host is up (0.000043s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http

Nmap done: 1 IP address (1 host up) scanned in 4.81 seconds
```

Step 2: Save Scan Output in XML Format

nmap 192.168.1.5 -oX scan.xml

```
(kali㉿kali)-[~]
└─$ nmap 192.168.10.3 -oX scan.xml

Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-31 11:19 -0500
Nmap scan report for 192.168.10.3
Host is up (0.000040s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http

Nmap done: 1 IP address (1 host up) scanned in 4.82 seconds

(kali㉿kali)-[~]
```

Step 3: Save Scan Output in All Formats (Most Used) nmap

192.168.1.5 -oA scan_result

```
(kali㉿kali)-[~]
$ nmap 192.168.10.3 -oA scan_result
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-31 11:20 -0500
Nmap scan report for 192.168.10.3
Host is up (0.000036s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http

Nmap done: 1 IP address (1 host up) scanned in 8.73 seconds
```

Step 4: Save Advanced Scan Results sudo

`nmap -A 192.168.1.5 -oA full_scan`

```
(kali㉿kali)-[~]
$ sudo nmap -A 192.168.10.3 -oA full_scan~
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-31 11:22 -0500
Nmap scan report for 192.168.10.3
Host is up (0.000058s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
80/tcp    open  http      Apache httpd 2.4.66 ((Debian))
|_http-server-header: Apache/2.4.66 (Debian)
|_http-title: Apache2 Debian Default Page: It works
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose
Running: Linux 5.X|6.X
OS CPE: cpe:/o:linux:linux_kernel:5 cpe:/o:linux:linux_kernel:6
OS details: Linux 5.0 - 6.2
Network Distance: 0 hops

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 16.43 seconds
```

Step 5: View Saved Results cat

`scan.txt`

```
(kali㉿kali)-[~]
$ cat scan.txt
# Nmap 7.98 scan initiated Sat Jan 31 11:18:12 2026 as: /usr/lib/nmap/nmap --privileged -oN scan.txt 192.168.10.3
Nmap scan report for 192.168.10.3
Host is up (0.000043s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http

# Nmap done at Sat Jan 31 11:18:17 2026 -- 1 IP address (1 host up) scanned in 4.81 seconds
```

7.INTERPRET RESULTS

Interpreting scan results is the process of analyzing the information collected during network scanning to understand security risks. After identifying open ports, services, operating systems, and vulnerabilities, the tester evaluates which findings are critical, moderate, or low risk. This step focuses on understanding how discovered vulnerabilities could impact confidentiality, integrity, and availability of systems. Not all vulnerabilities pose the same level of threat; therefore, interpretation helps prioritize remediation efforts. Proper risk interpretation is essential for making informed security decisions and preventing potential attacks.

8. DOCUMENT FINDINGS

This includes details such as identified hosts, open ports, detected services, operating systems, vulnerabilities, and associated risk levels. Documentation ensures that technical findings can be clearly communicated to stakeholders, system administrators, or management. Proper documentation is essential for compliance, auditing, remediation planning, and future reference. A wellwritten report serves as proof of assessment and guides corrective actions.

```
(kali㉿kali)-[~]  
$ nmap -A 192.168.10.3 -oA final_scan  
  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-31 11:30 -0500  
Nmap scan report for 192.168.10.3  
Host is up (0.000068s latency).  
Not shown: 999 filtered tcp ports (no-response)  
PORT      STATE SERVICE VERSION  
80/tcp    open  http      Apache httpd 2.4.66 ((Debian))  
|_http-server-header: Apache/2.4.66 (Debian)  
|_http-title: Apache2 Debian Default Page: It works  
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port  
Device type: general purpose  
Running: Linux 5.X|6.X  
OS CPE: cpe:/o:linux:linux_kernel:5 cpe:/o:linux:linux_kernel:6  
OS details: Linux 5.0 - 6.2  
Network Distance: 0 hops  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 12.55 seconds
```